



Spatial and volume patterns of an unmanaged submontane mixed forest in Central Europe: 160 years of spontaneous dynamics

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ARTICLE INFO

Article history:

Received 20 March 2011

Received in revised form 16 May 2011

Accepted 17 May 2011

Keywords:

Primeval forest dynamics

Volume and spatial pattern

Living and dead wood

Fagus sylvatica L.

Picea abies L. (Karst.)

Abies alba Mill.

ABSTRACT

The long-term development of stand characteristics and tree spatial patterns (TSP), their mutual relation, and linkage to site and tree species were studied in the Boubín primeval forest (protected since 1858). Surveys were carried out in 1851, 1961/1964, 1972, 1984/1989, 1996 and 2010 on one to six research plots sized 0.58–1.00 ha. To see how results from these surveys could be generalized, results were also compared with whole-area data sets (46 ha) from 1972 to 2010.

The proportion of *Abies alba* continually decreased, from 20% in 1851 to 2–3% in 2010. This decrease started no later than at the beginning of the 20th century. In contrast, the proportion of *Fagus sylvatica* slightly increased. The sum of dead and living wood volume was stable during the 159 year period, with deviations of only up to 5%. From 1961/64 to 2010, the number of living trees continually decreased, but the mean-tree volume and volume of dead wood increased. The distribution of dead wood always differed from the distribution of living trees.

A random TSP was always most common, which seems to be typical for this type of forest. This was true even when *A. alba* was gradually forced out from the stand structure. Nevertheless, the pronounced decrease of *A. alba* found during the 1961/64 survey compared with 1851 was reflected in a tendency towards clustering, as gaps from *A. alba* dieback gradually closed. In these gaps, competitive pressure was lower, resulting in more clustered distributions. Regular distribution was recorded only rarely. *F. sylvatica* maintained a random or clustered TSP over long periods, depending on site conditions. On water-affected plots, it had a consistently higher tendency towards clustering. The only step change in TSP occurred due to Hurricane Emma in 2008. Otherwise, the most marked TSP changes over time were found for *Picea abies*, in which there was a gradual trend from clustered to random, connected with a decreasing number of individuals.

The main reason for the decline of *A. alba* and the decreased number of living trees from 1961/64–2010 was the on-going effect of a high stock of red deer at the turn of the 19th and 20th centuries. The decreasing trend in the number of living trees showed no marked change, even 60 years after the game pressure was strongly reduced. However, the vertical structure became increasingly homogenized due to the long-term absence of severe abiotic and/or biotic disturbances.

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1. Introduction

Spatial and volume patterns of a forest are the result of many processes and factors, such as disturbance history, the variability of site conditions or competition between tree individuals, all operating at different scales of time and space. In most current studies,

evaluations of forest dynamics are inevitably based on one or a few snapshots in time, which do not allow detailed cause-effect relations to be examined, making it very difficult to connect the resulting stand structure to specific processes. This is even more so considering that one process may generate more types of structure and, on the other hand, one type of structure may be generated by multiple processes (Lepš, 1990). Because of these limitations, it is useful to study forest structure changes over long observational time series. Knowledge of the long-term development of a forest may serve to generate relevant hypotheses about the role of biological processes in this development (Aldrich et al., 2003; Dale, 1999). For this purpose, long-term studies of objects with a low number of overlapping variability factors are most suitable.

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The Boubín primeval forest is an ideal site for studying the development of stands. It has been under stringent protection since as early as 1858 (e.g. Vrška et al., in press), and is one of the oldest reserves in Europe (Welzholz and Johann, 2007). Natural processes are assumed to have been spontaneous, with no direct human impact. The first research into forest structure was already conducted in 1851, and we believe that this was the first complex survey of stand characteristics in an old growth forest ever made in the world. We compared research results from 1851, 1961/1964, 1972, 1984/1989, 1996 and 2010, obtaining unique information about the development of this primeval forest reserve in the heart of prehistorically colonized central Europe over nearly 160 years. Although the tree layer development has never been studied in Boubín in such detail, the reserve has served as a model example in construction of the developmental cycle concept of central European temperate forests (e.g. Leibundgut, 1993).

There are only a few works describing the long-term development of forest spatial and volume patterns and the impact of individual variability factors on these characteristics (e.g. Aldrich et al., 2003; Veska et al., 2009; Ward et al., 1996; Wolf, 2005). Based on these works, it can be hypothesised that the absence of extensive disturbances or direct human interference and the stability of site conditions at Boubín has allowed gradual evolution over the last 160 years, and reflects long-term trends (see Ward et al., 1996). However, the questions remain how the effects of Hurricane Emma, which marginally affected the locality on 1 March 2008, and the decline in *Abies alba* observed between 1972 and 1996 (Šamonil and Vrška, 2007b; Vrška et al., 2001) are reflected in this development. The impact of game animals in the first half of the 20th century is also difficult to quantify, due to the absence of relevant studies. We expect that game slowed down the growth of recruits, whose influence on the development of tree species pattern is considerable (Wolf, 2005), thus slowing down the development of the forest.

In this study, we explain (i) how the volume and spatial patterns of this unique primeval temperate forest actually changed, considering the possible *A. alba* decline over nearly 160 years, and (ii)

what are the relationships between the stand volume features, species representation, spatial patterns and site characteristics.

2. Material and methods

2.1. Study site

Our research took place in the Boubín primeval forest National Nature Reserve (hereinafter Boubín) in the Šumava Protected Landscape Area (PLA, Fig. 1). Prince Jan Adolf Schwarzenberg declared 144 hectares of this primeval forest exempt from management in 1858; however, a windstorm in 1870 resulted in disastrous wind breakages and a part of the reserve was severely disturbed. To prevent the possible spread of *Ips typographus*, the wind breakages were removed and the primeval forest area was reduced to its present 46 ha. After the Žofín virgin forest (Novohradské hory Mts.), Boubín is the second oldest forest reserve in the Czech Republic and one of the oldest in Europe (Welzholz and Johann, 2007). The primeval forest is located in the lower part of the north-eastern slope of Boubín Mt. at an altitude of 930–1110 m a.s.l. Three minor streams drain the Boubín locality into the Kaplický potok Brook.

Boubín lies on Bohemian Massif crystalline formed by primary schists, biotic and mica-schist gneisses. Lower elevation soils can be classified (Driessen et al., 2001; Michéli et al., 2006) as Endoskeletal Cambisols (Dystric) or Endoleptic Cambisols (Endoskeletal and Dystric); Umbric Podzols (Entic and Endoskeletal) occur at higher elevations and Arenic Gleysols (Humic and Eutric) up to Ombric Histosols occur in terrain depressions and along streams (Vrška et al., in press). Mean annual temperature is 5 °C and mean annual precipitation amount is 1300 mm (Tolasz et al., 2007). According to a floristico-phytocoenological classification (Braun-Blanquet, 1921; Ellenberg, 1996), most plant communities can be classified as *Calamagrosio-villosae Fagetum* Mykiška 1972 and *Calamagrostio villosae Piceetum* Hartman and Jahn 1967 (see Šamonil and Vrška, 2007b).

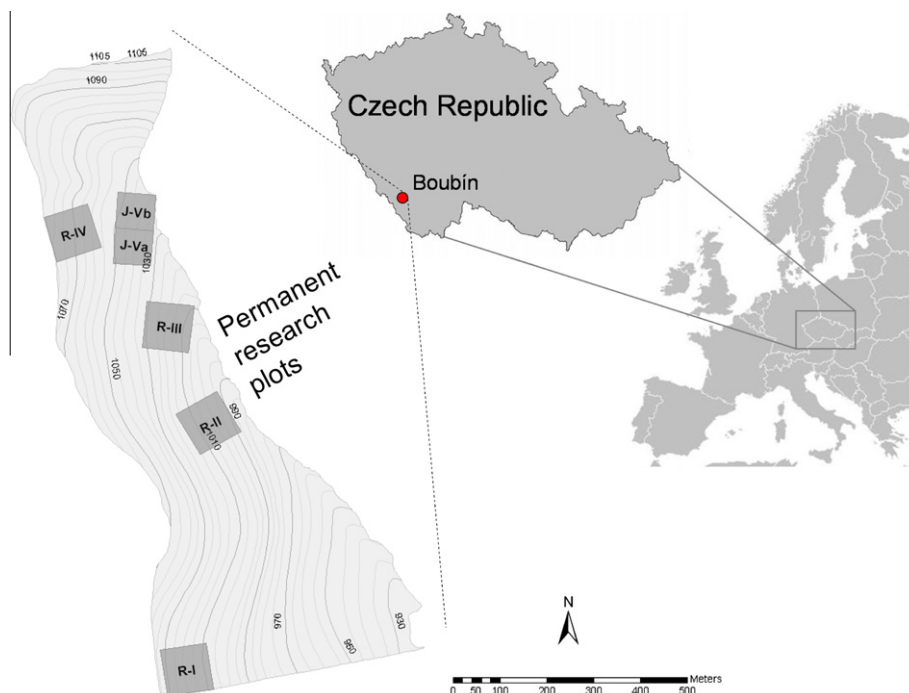


Fig. 1. Location of the Boubín primeval forest (hereinafter Boubín) in the Czech Republic. The image includes the core zone (46 ha, altitude 930–1100 m a.s.l.) with the layout of permanent research plots of John (J–Va, J–Vb; established in 1851) and Řehák (R–I, R–II, R–III, R–IV; measured in 1964).

Continuous primeval forests covering the Boubín Mt. were used for hunting as early as in the 17th century. The last wolf in the region was killed in 1756 and the population of red deer started to rapidly increase. The game stock peaked in 1764, when a liquidation program was commenced. At the turn of the 18th and 19th centuries, red deer in the region were rare, and the last animals were shot in 1827 (Kruml, 1964). In 1874, however, the primeval forest was included in a new red deer enclosure, and the re-introduced game stock reached another maximum in 1901 (Kruml, 1964). In spite of the fact that the red deer population dramatically decreased in the subsequent years, natural regeneration in the primeval forest was affected by red deer until World War II when the game enclosure ceased to exist (Průša, 1989). In 1966, the primeval forest was fenced, so the impact of game has been eliminated up to the present.

2.2. The history of research and data collection

The tree layer development in Boubín began to be monitored on several research plots in 1851 and across the whole primeval forest area in 1972 (Průša, 1985; Vrška et al., 2001; <http://www.pral-esy.cz>; Fig. 1 and Table 1).

- In 1840, head forester Josef John recorded the remnants of virgin forests in the Zatoň forest district. He decided to rescue the stands from felling and to use them for studying the dynamics of primeval forest dynamics. In 1847–1851, he established eight research plots along an altitudinal gradient on Boubín Mt. (1362 m a.s.l.), each sized one Lower-Austrian morgen (0.5755 ha). For each plot, a stem position map was plotted along with the number of trees and proportion of tree species, dead wood volume, mean-stem diameter and volume (Fig. 2, Nožička, 1958). As far as we know, this work is the first complex survey of stand characteristics ever conducted in a primeval forest worldwide. Unfortunately, the sketch map showing the distribution of plots in the field has been lost.
- In 1961 and 1964, four research plots across different forest sites and stages of forest development were delineated in Boubín, each 1 ha in size (I, II, III, IV). Parameters assessed on these plots were DBH and the height of all trees (by tree species, DBH ≥ 5 cm); the position of living and dead standing trees, stubs and lying trees was established by needle theodolite and a stem position map was plotted on a 1:200 scale (Fig. 2, Řehák, 1962, 1964). Two boundary stones were found that marked plot V established in 1851. Surveys of stand features were subsequently made of two possible variants of this plot (Va and Vb, Řehák, 1964).
- In 1972, stand characteristics were measured for the whole primeval forest and stem position maps were plotted at a scale of 1:1000 (DBH ≥ 10 cm, 46 ha), (Fig. 2). In addition to the

measurement of stem positions (accuracy ± 1 m), other parameters assessed included tree species, DBH, height and condition of individual trees (living standing, dead standing, wind throw, breakage, stub, or lying stem) (Průša, 1985, 1989).

- Measurements of stand characteristics were gradually repeated on Plots I–IV in 1984 and 1989 (Vaněk, 1990). Unfortunately, stem position maps were not preserved. Plots Va and Vb were not re-measured.
- Repeated measurements of stand characteristics for the whole primeval forest area were made in 1996, including the measurement of stem position and the construction of a stem position map (Vrška et al., 2001) (Fig. 2). The methodology was identical to that used in 1972 (Průša, 1985).
- We performed the most recent survey of the whole primeval forest area including the measurement of stem positions in 2010 (<http://www.pralesy.cz>).

2.3. Data analysis

In order to be able to study the development of stand characteristics and the development of spatial structure, we selected only comparable indicators from the corresponding plots (Table 1). These variables, obtained from the respective surveys and stem position maps, were as follows: volume of living and dead trees by species, number of living and dead trees by species and distribution according to DBH. We focused on analysing spatial data to study the distribution type for all trees and separately for individual species.

The original printed maps were digitalized and edited in the programme ArcGIS 9.1 (<http://www.esri.com>). Data for the research plots were excerpted from the whole-area surveys of 1972, 1996 and 2010 (Průša, 1985; Vrška et al., 2001; <http://www.pralesy.cz>). To make results more robust and clear, the data from individual surveys were averaged for plots I, II, III and IV measured in 1964 and for the Va and Vb variants of plot V established in 1851.

Tree volumes were determined identically during individual surveys and data sets from the volume tables of tree species (e.g. Anonymous, 1952) according to breast height diameter and fitted height. Volumes of standing or lying tree fragments were calculated as the sum of the volumes of approximation cones, following a mathematic model of stem form in the relevant species (Petráš 1989, 1990). Calculations of stem volumes were carried out in the PraleStat software (<http://www.pralestat.wz.cz>).

Trends in stand characteristic indicators were assessed using redundancy analysis (RDA) in the Canoco for Windows 4.5 environment (Lepš and Šmilauer, 2003; ter Braak and Šmilauer, 2002). The year of survey was used as the environmental variable, and locality code as covariable data (e.g. Šamonil and Vrška, 2007a, 2007b). This procedure allowed us to remove undesirable data

Table 1

Character of preserved data sets from the respective years of research. The data are separated into tabular dendrometrical data (stand characteristics were used to calculate changes in the volume distribution and diameter distribution of stands) and spatial data, with plotted geographical positions of all individuals in stem position maps (that were used to study spatial patterns); dead trees are divided to either standing (including stubs and breakage), and lying; ✓ – full evaluation, ✓* – evaluation without species discrimination, × – no evaluation. For survey details see Section 2.2.

Year of measurement	Research plot no.	Whole-area data	Dendrometrical data				Spatial data		
			Tables				Stem position maps		
			Living trees	Dead trees			Living trees	Dead trees	
				Standing	Lying			Standing	Lying
1851	V	×	✓	✓*	✓*	✓	✓*	✓*	✓*
1961/64	I, II, III, Va, Vb	×	✓	✓	×	✓	✓	×	×
1972	I, II, III, IV, Va, Vb	✓	✓	✓	✓	✓	✓	✓	✓
1984/89	I, II, III, IV	×	✓	✓	×	×	×	×	×
1996	I, II, III, IV, Va, Vb	✓	✓	✓	✓	✓	✓	✓	✓
2010	I, II, III, IV, Va, Vb	✓	✓	✓	✓	✓	✓	✓	✓

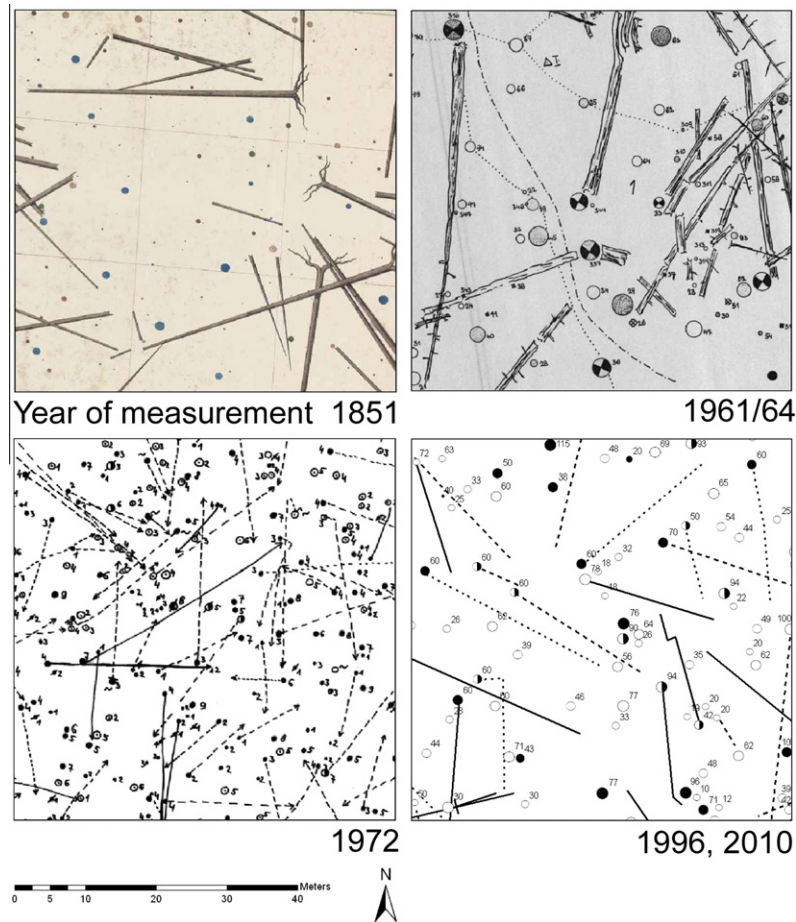


Fig. 2. Examples of stem position maps for research plot V from individual years of research – 1851, 1961/64, 1972, 1996 and 2010. Stem positions were used to study spatial relations on all research plots and to obtain stand characteristics.

variability resulting from differences among the plots, and enabled us to evaluate only changes on individual plots. The significance of the year of survey was tested by a Monte Carlo permutation test with 4999 permutations.

Ripley's function $K(r)$ (Ripley, 1977) was used to evaluate the tree spatial patterns (TSP). This function is defined as $K(r) = \lambda - 1E$, where λ is density (in this case the number of trees/stand area) and E is the number of trees within distance r of an arbitrary tree. Under the assumption of a completely random process, $K(r) = \tau\tau r^2$, $r \geq 0$. Under regularity, $K(r)$ tends to be less than $\tau\tau r^2$, whereas under clustering $K(r)$ tends to be greater than $\tau\tau r^2$. To investigate the spatial relation between species, we used the bivariate function $K_{ij}(r)$, which is the extended $K(r)$ function to multitype point patterns. Edge effects were corrected using Ripley's isotropic correction (Ripley, 1988). Monte Carlo simulations were conducted to test the null hypothesis of a completely random point process (Besag and Diggle, 1977). We used 39 simulations of this process to obtain pointwise critical envelopes for $K(r)$. The significance level of this pointwise Monte Carlo test was 0.05. The null hypothesis of complete spatial randomness can be rejected at fixed r if (1) the value of $K(r)$ is larger than the value of the pointwise upper envelope, and thus the distribution is clustered, or (2) the value of $K(r)$ is smaller than the value of the pointwise lower envelope, and thus the distribution is regular. All spatial analyses were conducted using the package "spatstat" (Baddeley and Turner, 2005) in the statistics software R (R Development Core Team, 2006). Patterns with fewer than 10 stems in a given plot were too rare to display any significant spatial structure. Applying the method described by Wolf (2005), we computed the degree of change in the spatial patterns.

Using RDA analysis, we tested the relation between stand characteristics and the type of spatial pattern for all species, in individual trees and the pattern of interspecific links. The year of survey was used as covariable data. Similarly as in the analysis of stand characteristic trends over time, statistical testing was performed in the programme Canoco for Windows (Lepš and Šmilauer, 2003; ter Braak and Šmilauer, 2002).

Extensive stem position maps (46 ha) from the shorter period 1972–2010 were used to describe the distribution of basic stand structural features and tree species composition. Comparing the observed distributions with the mean values estimated on fixed research plots, we could evaluate the representativeness of these mean estimates and the trends over time observed in the data. A computer-simulated placement of 1ha square sample plots was performed by a moving window method (Král et al., 2010a,b). In this manner, a 1 ha plot was virtually placed in a 10 m grid in each stem map measured in 1972, 1996 and 2010. For each simulated plot, the four basic stand characteristics (hectare density, volume of living trees, volume of dead trees and total stand volume per hectare) were calculated in total and then separately for the three principal tree species – *Picea abies*, *Fagus sylvatica* and *A. alba*. In order to reduce edge effects, we excluded a 50 m buffer around the study site.

3. Results

3.1. Development of stand characteristics

The year of survey was highly significant ($p < 0.001$, $F = 21.98$, Fig. 3), explaining 55.0% of the data variability. A considerable

decrease in *A. alba* over time was evident in all evaluated indicators. The trend was most clear on plot V, where the original proportion of this species 19.8% in 1851 (179 m³/ha) dropped to a mere 2.6% in 2010 (19.3 m³/ha, Fig. 4). On plots I–IV, this species had an average proportion of only 2.1% (14 m³/ha) in 2010, compared to 11.7% (101 m³/ha) in 1961/64. As expected, the dead wood volume of the species increased with the long-term decrease of the volume of living trees. The relevant proportion of *A. alba* in dead wood in 1961/64, however, demonstrates that the species decline had started long before. This is further supported by the trend in individual stem decay, which can be seen in detail between 1961/64 and 2010. After nearly 50 years, 58% of originally lying *A. alba* stems remained non-decomposed.

In general, the species distribution of dead wood largely differed from the distribution of living wood. The predominant species was *P. abies*, whose volume considerably increased over time, while in contrast the proportion of *F. sylvatica* did not exceed 31%. The rate of *F. sylvatica* decay was significantly higher than in the coniferous species: only 27% of originally lying stems were not completely decomposed between 1961/64 and 2010. The total volume of dead wood as well as the number of dead individuals significantly increased over the last 50 years, reaching those values recorded in 1851 (Fig. 4).

The sum of living and dead wood volumes was surprisingly stable over the nearly 160 years, ranging from 1170 to 1287 m³/ha (incomplete data were not taken into account). The total volume of living and dead stems on plots I–IV ranged from 959 to 1009 m³/ha. The lower total timber volume on plots I–IV resulted from a higher proportion of *F. sylvatica* (46–51%, Fig. 4), which, unlike *P. abies*, does not reach tree heights over 50 m and DBH over 120 cm in Boubín. The situation on plot V was different. This plot has been dominated by *P. abies* (65–75%) since 1851. However, the proportion of *F. sylvatica* steadily increased, from 15.3% in 1851 to 25.3% in 2010.

The number of living trees generally decreased, in particular *F. sylvatica* and *P. abies* individuals in the lowest diameter classes (up to DBH = 35 cm) from 1961/64 to 2010. An opposite trend, i.e. an increasing number of individuals over time, was observed in the higher diameter classes of *F. sylvatica* and *P. abies* (DBH ≥ 65 cm, Figs. 3, 5 and 6). Thus, the mean-tree volume of *P. abies* and *F. sylvatica* increased over time (Fig. 6). Nevertheless, this increase did not prevent a general decrease in the volume of living trees by 2010.

Over 160 years, though, the situation was more complex. While the number of *A. alba* (and *F. sylvatica*) individuals decreased from 43 (250) per hectare in 1851 to 4 (70) per ha, the number of *P. abies* individuals first increased from 82 per ha in 1851 to 354 per ha in 1961/1964 and to 359 per ha in 1972, but then began a decrease to only 164 individuals per hectare in 2010 (plot V). A more distinctive peak in the number of individuals in the lowest diameter class was also observed in 1961/1964 for dead wood. Within a decade, the number of individuals in this class had considerably decreased (Fig. 6).

3.2. Development of spatial patterns

The spatial pattern analysis revealed that the development of individual plots was not consistent. Differences existed among the plots not only in the type of spatial distribution but also in dynamics (Table 2). Considering the spatial structure (clustered, random, or regular), the pattern was always random at distances up to 5–10 m, and both random and clustered at over 10 m. In just one case (patterns of *F. sylvatica* vs. *P. abies*) were all types of spatial structure present over time. The regular distribution type was only found between these species in approximately 20% of cases.

In the majority of cases, no significant relation was found between the type of spatial distribution and stand characteristics. Only the *F. sylvatica* spatial pattern showed a significant relation ($p = 0.002$, $F = 8.22$) (Fig. 7). This species remained clustered for nearly 50 years, with frequency and proportional volume being markedly lower than for *P. abies*. In contrast, the random spatial pattern was predominant on plots with high volumes and numbers of *F. sylvatica* and at sites with a higher volume of dead wood.

Compared with *F. sylvatica*, the *P. abies* pattern was more variable over time. However, random distribution was still predominant in this species (64%), with the remaining 36% clustered. Along with the decreasing number of stems, plots I and III exhibited a slow decrease in clustering and a transition to randomness. On plots II, IV and Va, the pattern of *P. abies* did not show any particular change, whereas on plot Vb there was a distinctive step change between the years 1996 and 2010.

In most cases, spatial relations among *A. alba* individuals could not be established due to the low numbers of this species. Where it could be determined, the *A. alba* TSP was always random.

The random pattern also predominated (86% of cases) in spatial relations among all living tree individuals. While the spatial

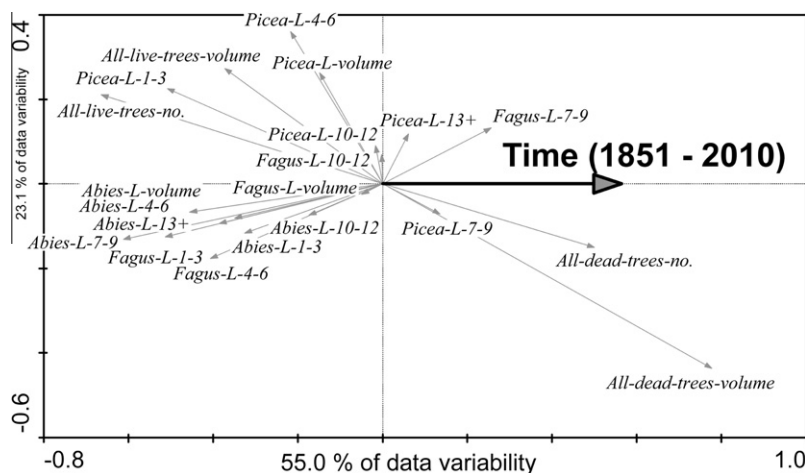


Fig. 3. Development of stand characteristics in the period from 1851 to 2010. RDA analysis; the year of survey was used as the environmental variable, and research plot code as covariable data; the variable explains 55.0% of the data variability; the vertical axis is the first non-canonical axis; for redundancy analysis adjustment, see Section 2.3. Non-averaged data from individual plots in respective years were used for analysis. Data from plots V, Va and Vb were converted to 1 ha. Abbreviated names: *Fagus* – *Fagus sylvatica*, *Picea* – *Picea abies*, *Abies* – *Abies alba*, L – living trees, volume in m³/ha, diameter at breast height (DBH) aggregated to 30 cm-classes (e.g. 4–6 – DBH = 35–65 cm).

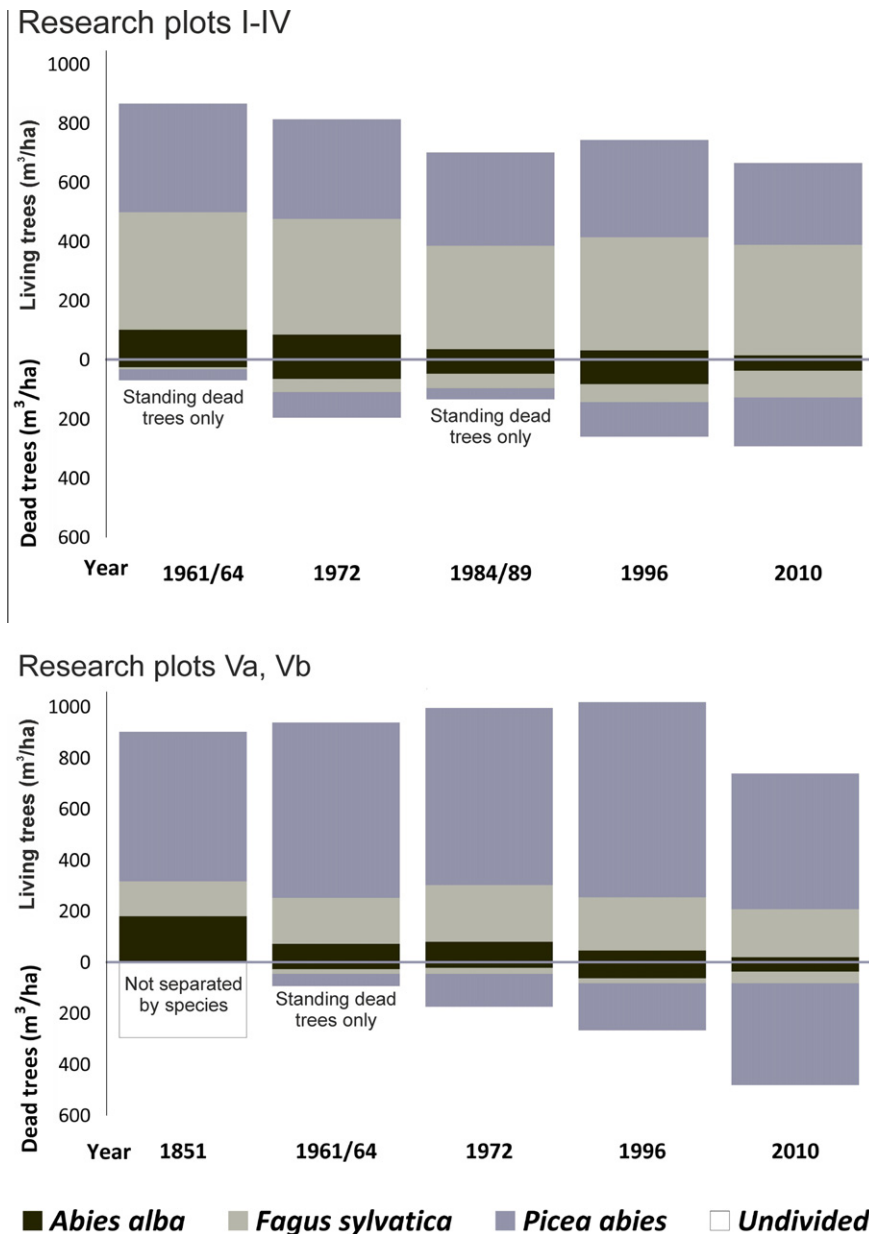


Fig. 4. Volumes of living and dead trees on the research plots in respective years according to tree species. Average data from plots I–IV and Va–Vb converted to 1 ha are shown. The volume of dead wood in 1851 is not separated by species as the information was missing in the original records available; for the same reason, the volume of dead trees in 1961/64 and 1984/89 is only represented by dead standing trees. For calculation details see Section 2.3.

pattern showed continual development from 1961/1964 to 2010, more distinct changes in the spatial distribution of tree species occurred between 1851 and 1961/1964. In this period, *F. sylvatica* showed a trend from random to nearly regular distribution and then towards clustered. The degree of change ΔK (14.0 m) between 1961/64 and the last observation in 2010 was not significantly correlated with the number of trees per hectare at the beginning of the study (Pearson correlation coefficient = 0.506, $p = 0.305$, $n = 6$).

3.3. Local variability of stand structural features

Analysis of the whole-area data (Fig. 8, 46 ha) revealed considerable local variability in stand characteristics. The median as well as variance of most investigated variables changed significantly from 1972 to 2010. Since 1972, the number of all living trees and volume of *A. alba* have significantly decreased, whereas the volume of dead wood increased; volumes of *F. sylvatica* and *P. abies* and the

total volume changed less dynamically. The changes observed in plots I–IV and Va–Vb coincided well with the whole-area development trends.

Mean stand characteristics of plots I–IV also corresponded very well with the mean values from the whole-area, and thus can be described as representative. Plots Va and Vb were different, however. Although developments in these plots followed the overall trends very well, individual variable values were different. Variations were especially apparent in the volume of *P. abies* and *F. sylvatica*, as well as the volume of dead trees in 2010 (Fig. 8).

4. Discussion

4.1. Generalizability of the results from the research plots

Although the use of plots sized 0.5–1.0 ha is considered standard in current scientific literature (e.g. Arévalo and

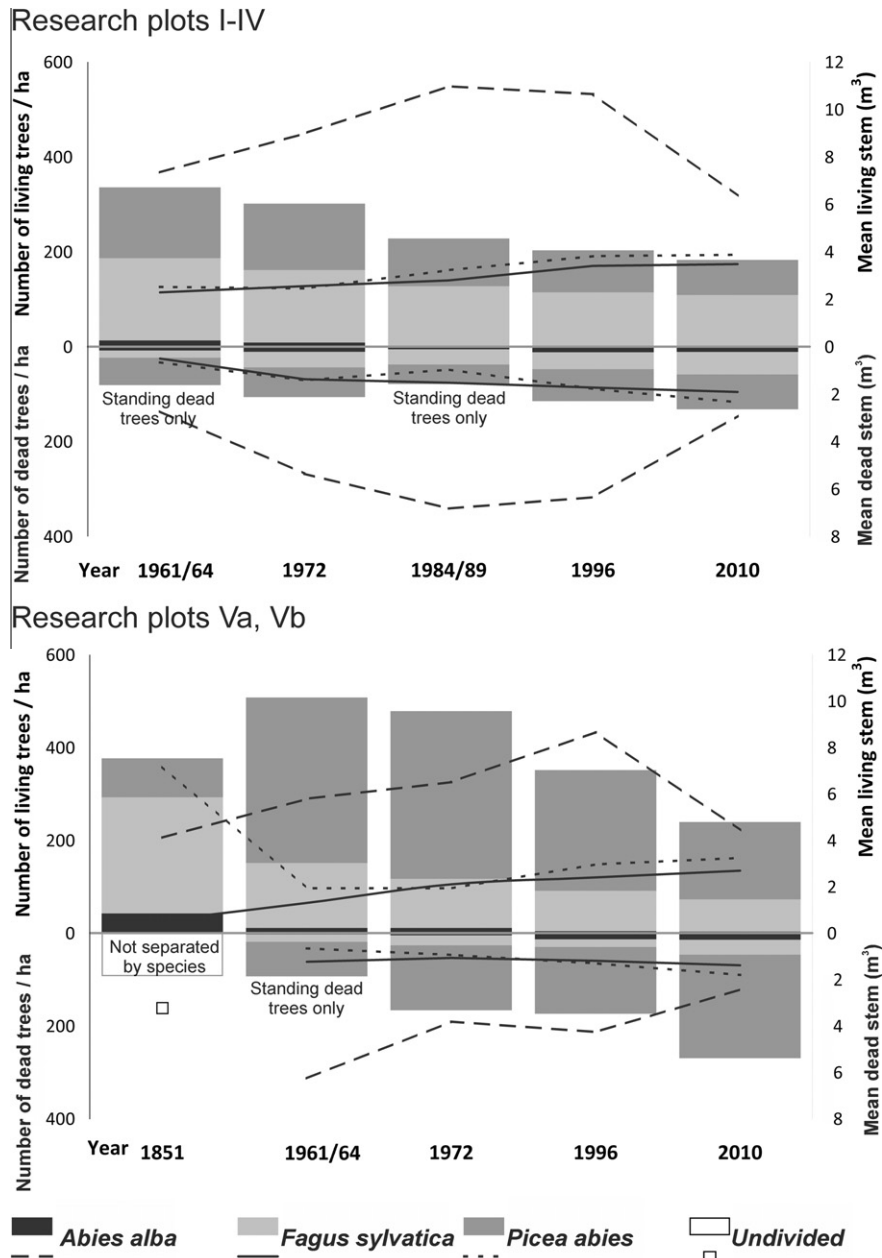


Fig. 5. Development of number of living and dead trees on the research plots in respective years, according to tree species (histogram) in relation to volume of mean living and dead tree (curves), respectively. Data were averages from plots I–IV and Va–Vb converted to 1 ha. The values of dead trees in 1851 were not separated by tree species as this information was missing in the available records; for the same reason, the volume of dead wood in 1961/64 and 1984/89 is only represented by dead standing trees.

Fernández-Palacios, 2003; Fajardo et al., 2006; Jaworski and Paluch, 2001; Salas et al., 2006; Wolf, 2005), our previous analyses made in natural temperate forests have demonstrated the limitations of such studies in comparing major stand structural features using a limited number of sample plots. Chave et al. (2001, 2003) arrived at similar conclusions in neotropical forests. In the publication Král et al. (2010a), we calculated the need for 6–18 plots sized 1 ha in beech-dominated natural forests (depending on the estimated variable) in order to estimate main stand variables to within 20% ($\pm 10\%$) of the mean with 95% confidence. With this number of plots, the local vegetation variability is sufficiently taken into account. However, this requirement could not be met in our current study. Though whole-area data covering 46 ha were available from research conducted in 1972, 1996 and 2010, corresponding data from older surveys, which were of key significance for studying

the long-term dynamics, were lacking. However, the fact that we also had whole-area data enabled us to assess how the results from our research plots reflected more general patterns.

While the mean values of stand characteristics and trends for plots I–IV coincided with the whole-area data, plots Va and Vb were consistent “only” with the trends. The markedly high proportion of *P. abies* and related high volume and number of individuals are caused by these plots being more affected by water and the affinity of *P. abies* for such habitats. Thus, while plots I–IV well represent the whole-area primeval forest development, plot V rather represents water-affected habitats, which are relatively rare in Boubín (according to Vrška et al. (2001) occupying 14.28 ha). In plots Va and Vb, the volume decrease of living trees and volume increase of dead wood was also noticeable between 1996 and 2010. These changes were partly attributable to the impact of Hurricane

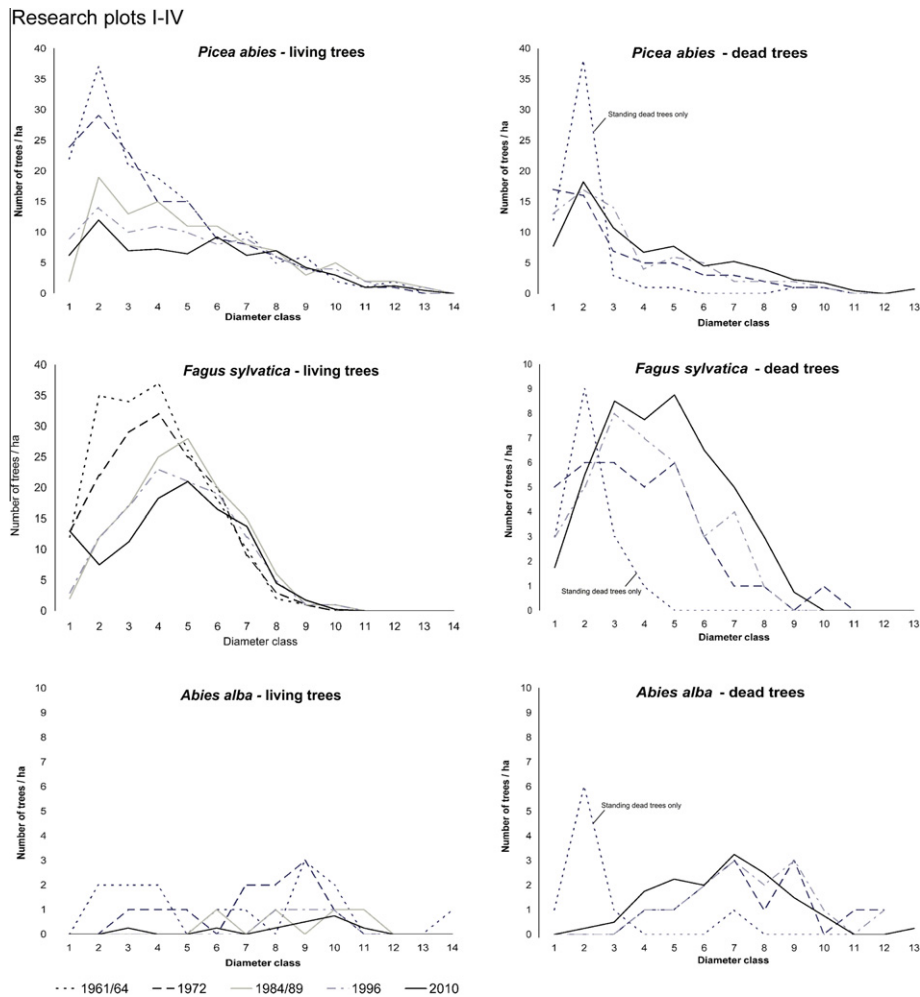


Fig. 6. Distribution of the diameters at breast height (DBH) of living and dead trees on the research plots for individual tree species according to year of measurement. Data are averages from plots I–IV and Va–Vb and converted to 1 ha. Plots Va and Vb are missing the development curve of the DBH distribution of tree species in dead trees, as a separation by tree species was missing in the original record. Distribution of DBH of dead trees recorded only standing dead trees in 1961/1964.

Emma in 2008. Apart from whole-area trends, however, the results from these research plots also clearly reflect local changes in the stands connected with changes of forest development stages and local site conditions (Korpel, 1995; Král et al., 2010b; Šamonil and Vrška, 2007a). This is the main reason for occasional differences in individual research plot dynamics, which did not reach the variability range of the tree layer. Similar differences in the development of research plots over 60 years were found also by Aldrich et al. (2003).

4.2. Stand characteristics

The most obvious trend in the development of the Boubín tree layer was the decrease of *A. alba*, which had commenced by the early 20th century at the latest. This decrease in fir was also observed at other localities in central Europe already before World War II (e.g. Chmelař, 1958, 1959; Málek 1981, 1983; Šnobl and Künzel, 1934).

A number of current studies have associated the decline of *A. alba* in natural forests with historic human management practices (Diaci et al., 2010; Kozáková et al., 2011; Šamonil and Vrška, 2007a; Vrška et al., 2009). According to these authors, cattle grazing and litter raking in forests promoted *A. alba* indirectly, especially at the expense of *F. sylvatica*. On the other hand, the subsequent end of these activities, after which the forests were

allowed to undergo spontaneous evolution and herbivorous even-toed ungulates (namely *Cervus cervus*) returned, are considered to have led to increased browsing of the regenerating *A. alba* and to have limited its further spread. However, this “management” theory can only be applied to Boubín to a limited extent. The human impact in the region can primarily be seen in the control of red deer numbers (Ammer, 1996; Klopčic et al., 2010; Motta, 1996).

In our opinion, the period between years 1764 and 1874, with relatively low game pressure, was favourable for the development of *A. alba* in Boubín, as was recorded in 1851. The plot data from one of the (unfortunately not retraceable) research plots from 1851 even mentioned a density of 130 trees of *A. alba* per ha. The loss of trees dying in the main level was not compensated for after 1874 when game stock was high, resulting in a greater opening of stands. Such conditions were favourable for *F. sylvatica*, whose advance growth successfully competed with *A. alba*, which could not take advantage of its capability for slow growth in the shade. We do not consider site aspect or insolation intensity (Heuze et al., 2005) to be important factors in the competition of *A. alba* with *F. sylvatica* and *P. abies*, as the entire locality is situated on a shady NE slope in a lower section of the valley.

Wildlife might not have been the only causal factor in the decrease of the *A. alba* population. Published works repeatedly discuss the impact of air pollution. Most recently, Elling et al.

Research plots Va, Vb

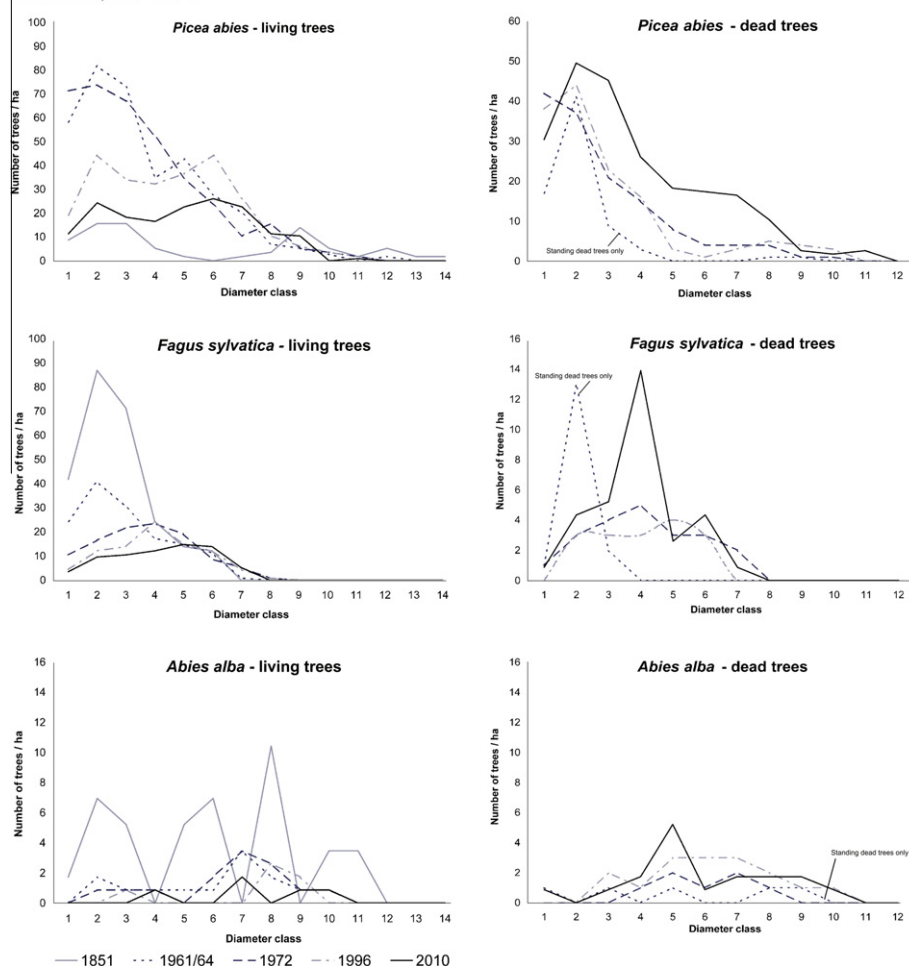


Fig. 6 (continued)

(2009) convincingly described the crucial effect of SO_2 emissions on the worsening health condition of *A. alba* in Germany from the 1960s to 1980s, and consider an atmospheric concentration of $10 \mu\text{g SO}_2 \text{ m}^{-3}$ to be critical. Nevertheless, the decline of *A. alba* in Boubín had apparently started long before the research conducted in 1961/64. A substantial proportion of the abundant *A. alba* population from 1851 was not found in the 1960s even among dead trees, even though at least half of them should have remained un-decomposed for 50 years. This is why we believe that the rise of post-war heavy industry might have catalyzed the dieback of *A. alba* but was not its primary cause. Moreover, we can presume from the trends of SO_2 emissions in the Czech Republic since 1850 (Kopáček and Veselý, 2005; Kopáček et al., 2001) and from the relation between recent atmospheric SO_2 concentrations and SO_2 emissions (e.g. Veselý et al., 1998) that the critical mean annual concentration of $10 \mu\text{g SO}_2 \text{ m}^{-3}$ was not exceeded at Boubín before 1961, as its geographical location would have only allowed for long-distance air pollution transport. The “air pollution” theory is therefore considered secondary, similarly as a possible increase in *F. sylvatica* competitiveness due to climate change (cf. Elling et al., 2009). We consider the main reason for the decrease of *A. alba* to be the long-term impact of a high red deer stock. In fact, despite the present fencing of Boubín, we do not expect a considerable recovery of *A. alba* in the future. On the contrary, synergy from the combined effect of all the above-mentioned factors could further increase the future proportion of *F. sylvatica*, which currently has advance growths below the recordable DBH limit.

The decreased number of individuals and the increased mean-stem volume for all species in the period 1961/64–2010 was surprising. This trend, especially visible on plots I–IV, was also clearly reflected in the whole-area data (Řehák, 1962, 1964; Vrška et al., 2001, in press; <http://www.pralesy.cz>). The trends in diameter distribution described above can be interpreted as a result of individual selection due to competition, related to the increase in volume and ageing, hence to the ageing of the whole forest segment (Král et al., 2010b). This development is partly in contrast to the traditional concept of alternating development stages in the cycle of temperate forests (Korpeľ, 1995; Král et al., 2010b; Leibundgut, 1993; Otto, 1994). With a long-term absence of severe disturbances in natural forests (as in Boubín), it is expected that a variegated mosaic of developmental stages will be established with patches sized 200 and 700 m^2 (Král et al., 2010a,b; Leibundgut, 1959, 1993; Mayer et al., 1987). Dynamic changes should occur at the level of individual patches, while at the locality level the concept assumes greater self-regulation. In contrast, large-scale changes are generally associated with succession (*sensu lato*) following natural or anthropogenic disturbances (e.g. Szwagrzyk and Szwagrzyk, 2001). This is the case in Boubín, where large-scale changes associated with game pressure at the turn of the 19th and 20th centuries are still being reflected in the forest structure – similar to the inertial effect of a large-area disturbance.

This game pressure disrupted the recruitment of individuals in the stand structure, which led to vertical simplification and large-scale homogenization of the stand structure. The formerly

Table 2

Changes of tree spatial pattern type on the research plots over time. The relations were evaluated for all species, individual species and the structure of relations among species for a distance gradient of 0–30 m. Numerals next to species names represent numbers of individuals per hectare. Spatial data from available stem position maps were used for calculating Rypley's function $K(r)$. For details on the calculation of spatial relations see Section 2.3.

Research plot	Year of measurement	All living trees	<i>Picea abies</i>	<i>Fagus sylvatica</i>	<i>Abies alba</i>	<i>Abies</i> vs. <i>Picea</i>	<i>Abies</i> vs. <i>Fagus</i>	<i>Fagus</i> vs. <i>Picea</i>
I	1961/64		59	186	21			
	1972		71	169	17			
	1996		51	129	9 x			
	2010		45	142	7 x			
II	1961/64		91	197	5 x	x	x	
	1972		86	173	4 x			
	1996		68	132	1 x	x	x	
	2010		51	124	0 x	x	x	
III	1961/64		323	161	14			
	1972		262	131	9 x			
	1996		143	85	7 x	x	x	
	2010		120	79	1 x	x	x	
IV	1961/64		114	151	15			
	1972		132	139	8 x			
	1996		83	104	1 x	x	x	
	2010		70	86	1 x	x	x	
V	1851		82	250	43			
	Va	1961/64	261	198	17			
		1972	281	148	16 x			
		1996	184	118	7 x	x	x	
		2010	137	94	7 x	x	x	
	Vb	1961/64	448	82	7 x	x	x	
		1972	436	64	9 x			
		1996	332	56	3 x	x	x	
		2010	191	45	2 x	x	x	

0 15 30 m

clustered distribution

random distribution

regular distribution

x low data

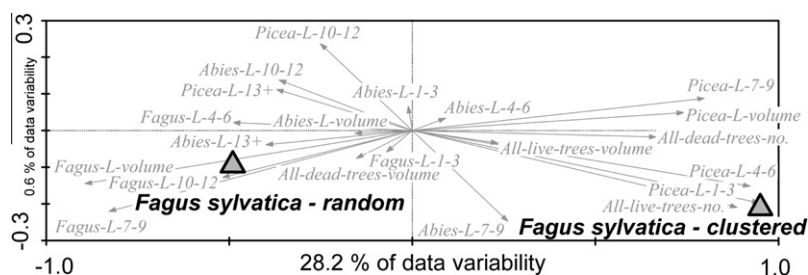


Fig. 7. Relation between the stand characteristics and tree spatial pattern (TSP) for *Fagus sylvatica*. RDA analysis; the year of survey was used as covariable data; type of TSP explains 28.8% of data variability; triangles represent centres of gravity of all research plots in individual years of measurement belong to random resp. clustered TSP; for redundancy analysis adjustment and the procedure for assessment of spatial relations see Section 2.3. Abbreviations are identical as in Fig. 3.

suppressed advance growths began to grow after the cessation of the game preserve; however, these did not reach the 10 cm DBH limit even by 2010 (44 years after fencing). Thus, we assume that the survey in 2010 recorded a situation where the number of living individuals in Boubín is at a minimum, and we predict increases in the future. The slowly increasing radial growth of trees is also reflected in the “generation” of individuals recorded in 1961/1964 in the lowest diameter classes, which is likely to have already originated from the period before the establishment of the game preserve in 1874. The vertical structure homogenization also increased due to the long-term absence of severe abiotic and/or biotic disturbances.

The distribution of dead wood is dominated by *P. abies* and *A. alba*, whose decay lasts much longer than that of *F. sylvatica*. While

reported *F. sylvatica* decay times do not exceed 50–60 years in a number of studies (Kraigher et al., 2002; Lombardi et al., 2008; Müller-Using and Bartsch, 2009; Šamonil et al., 2009, 2010), the decay time of *P. abies* generally exceeds 100 years (e.g. Storaunet and Rolstad, 2002). Our results confirm these findings. The distinctly different structure and especially dynamics of dead wood compared with living wood leads us to ask whether such differences should not be taken into account in the construction of forest development models (e.g. Král et al., 2010b).

4.3. Spatial patterns

Our study demonstrated the development of TSP in a spruce-fir-beech primeval forest over almost 160 years. Similar studies

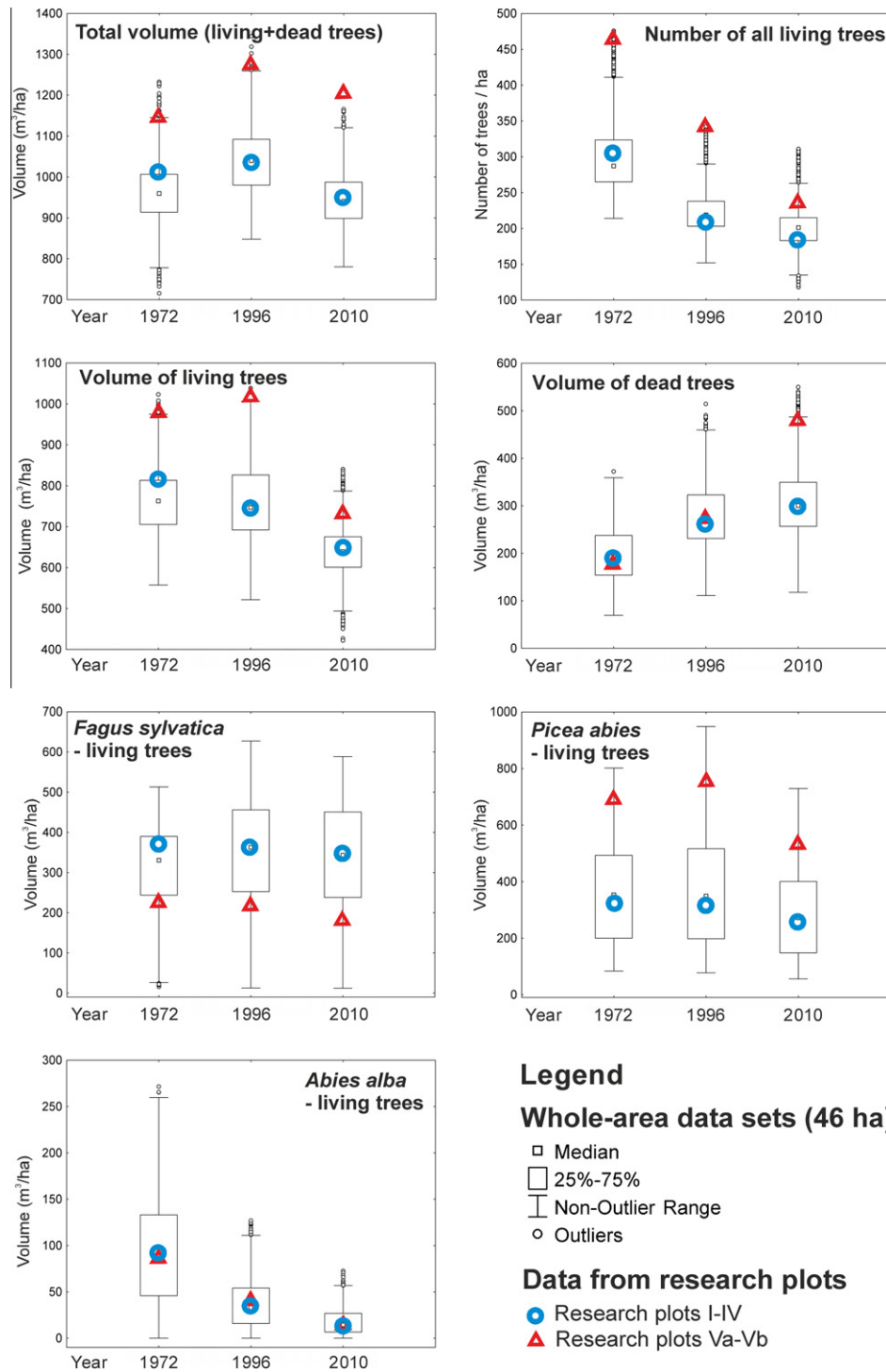


Fig. 8. Comparison of average per hectare values of stand characteristics between permanent research plots and whole-area data sets ($n = 2845$) by year of measurement. The circles and triangles represent values of a particular stand characteristic (number of all living trees, total volume of living and dead trees, volume of living trees, volume of dead trees, volume of living trees of *Fagus sylvatica*, *Picea abies*, and *Abies alba*) from plots I–IV and Va–Vb, respectively, in comparison with basic statistics of the whole-area data sets according to year of research. Data from plots I–IV and Va–Vb are averaged and converted to 1 ha. For details see Section 2.3.

are rare because relevant data are missing. Wolf (2005) found a considerable shift from the regular to random tree distribution within a distance of about 10 m over five decades, and explained this phenomenon by direct human influence in the past. However, forest dynamics in Boubín were affected by humans only indirectly through the regulation of the red deer stock, and the significance of this is not unambiguously clear. Changes observed between 1961/64 and 2010 rather correspond with the results of Aldrich et al.

(2003), who found high TSP stability. Long-term stability was seen in Boubín for the spatial patterns of *A. alba* (in 100% of cases), *F. sylvatica* (74%) as well as of all living trees (86%). *P. abies*, however, showed a more frequent gradual shift from the clustered (36%) to random TSP. This suggests that with the absence of extensive disturbances, the pattern of natural forests is a relatively robust characteristic. Nevertheless, severe disturbances can result in a step change in the development of TSP, as demonstrated by the impact

of Hurricane Emma in 2008 on plot Vb. On the other hand, with continual, gradual stand evolution, the tree spatial pattern may remain stable for decades, even when some main species are slowly forced out of the stand structure. For example, Aldrich et al. (2003) described a stable pattern type in spite of the fact that some shade-intolerant and mid-intolerant species, mainly *Fraxinus americana* and *Quercus rubra* (Parker et al., 1985) showed a significant decline. The decline of *A. alba* in Boubín was similar. Nevertheless, the decrease in the proportion of *A. alba* between 1851 and 1961/64 was distinctive enough (from 19.8% to 7.5%) that it was reflected in the *F. sylvatica* tree spatial pattern (and partly in *P. abies* as well) as a shift from random distribution in 1851 to clustered in later measurements. We explain this shift by the closing of gaps arisen after the dieback of *A. alba* individuals and by the temporary mitigation of tight competitive relations. One question is how the stand texture will continue to develop after the closing of gaps. A possible option is a return to the highly random pattern.

The results of our study may contribute to a discussion about the definition of TSP form in spruce-fir-beech primeval mountain forests. The predominant random spatial pattern – both cumulative for all tree species as well as for individual species – appears to be typical for this forest type, as Szwagrzyk and Czerwczak (1993) found in the beech and spruce-fir forests of the West Carpathians. However, the clustered spatial pattern often occurs for *P. abies* and *F. sylvatica* on water-affected sites.

4.4. Relation between stand characteristics and spatial pattern

The most often studied stand characteristic associated with the tree spatial pattern is stand density. Wolf (2005) concluded that areas with a higher tree density change less in pattern over time because they have a lower number of recruits. In our study, however, this relation was not significant ($r = 0.506$; $p = 0.305$). This may be due to the limited increase in the number of recruits on our plots. We also assume that the processes of tree spatial pattern changes significantly depend on the ecological requirements of particular species. This is why evaluating the dependence of TSP on stand characteristics should include the TSP analyses of individual species rather than a comprehensive analysis of all populations.

In their studies from east-central Indiana, Aldrich et al. (2003) and Ward et al. (1996) showed that for light-demanding species, a decreased number of individuals is usually accompanied by a decreasing trend toward clustering, while shade-tolerant species show the opposite trend. Results of our research are more or less in agreement with these conclusions. The decrease in the number of the more light-demanding spruce individuals from 1961/64 to 2010 is accompanied by decreased clustering and a shift to the random TSP. In contrast, clustering was observed to increase at the increased density of individuals in diameter class 1, as a consequence of gap regeneration (plot IV). The TSP dependence mentioned above was interrupted by the hurricane of 2008, when the number of individuals decreased and clustering increased (plot Vb). This clustering resulted from disruption of the former stand structure by wind, not from the occupation of gaps by recruits. The development of the tree spatial pattern in *F. sylvatica* (favouring shade – Ellenberg et al., 1992) was variable on the studied plots, and there was no specific relation between the density of individuals and changes in the tree spatial pattern.

Acknowledgements

The authors would like to thank their colleagues from the “Blue Cat research team” for field data measurement. They would also like to thank all anonymous reviewers, as their comments and suggestions considerably improved the quality of the paper. The research was supported by the Czech Science Foundation (Project

Nos. P504/10/2018,P504/11/2301), the Czech Ministry of the Environment (Project No. MSM 6293359101).

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